

Summary

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Useful formulas: $(1) \sum_{k=1}^n C = n \cdot C; \quad (2) \sum_{k=1}^n k = \frac{n(n+1)}{2}; \quad (3)$

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6} \quad (3) \sum_{k=1}^n k^3 = \frac{n^2(n+1)^2}{4}$$

Useful example of application of rules for sums:

$$\sum_{k=1}^n (ak + b) = \sum_{k=1}^n ak + \sum_{k=1}^n b = a \sum_{k=1}^n k + \sum_{k=1}^n b = a \frac{n(n+1)}{2} + nb, \text{ where } a \text{ and } b \text{ constants.}$$

SUMMARY of Riemann Sums:

Riemann sum: $\sum_{k=1}^n f(x_k^*) \Delta x,$

Right Riemann sum: $x_k^* = x_k;$ Left Riemann sum: $x_k^* = x_{k-1};$ Midpoint

Riemann sum: $x_k^* = \frac{x_{k-1} + x_k}{2}.$

Width of each of n rectangles on the interval $[a, b]: \quad \Delta x = \frac{b-a}{n}$

Grid points for interval $[a, b]:$

$$x_0 = a,$$

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$$x_{k-1} = a + (k-1)\Delta x = a + (k-1)\frac{b-a}{n},$$

$$x_k = a + k\Delta x = a + k\frac{b-a}{n},$$

Summary

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$$x_n = a + n\Delta x = a + n\frac{b-a}{n} = b$$